

EveNet for κ_λ Density-Ratio Learning in $HH \rightarrow b\bar{b}\tau^+\tau^-$: Pre-training Improves Discrimination and Ratio Shape, but Not Data Efficiency

Working note — $b\bar{b}\tau^+\tau^-$ κ_λ NSBI project

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Gates G0 / G0.5 of `EVENET_FEASIBILITY_NOTE`; 90-run sweep, branch `fm`

Abstract

We test whether the pre-trained event-level foundation model (FM) EveNet [1] improves the *data efficiency* of density-ratio learning for a Higgs self-coupling (κ_λ) measurement by neural simulation-based inference (NSBI), on CMS full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ events. Three arms — fine-tuned, frozen-backbone and from-scratch — identical in data, architecture and head and differing *only* in initialisation, were trained at six training fractions $\times$ five seeds (90 runs) and evaluated on one fixed held-out split against two axes fixed before unblinding: discrimination (weighted area under the receiver-operating characteristic curve, AUC) and ratio calibration (integral and shape closure). Three results. **(i) Discrimination: pre-training wins decisively** — $+0.014$ to $+0.017$ AUC at every fraction $\geq 1\%$, worth $\approx 3\text{--}4.5\times$ the training data, with both pre-trained arms exceeding a feature-level gradient-boosted decision tree (BDT) ceiling. **(ii) Calibration quality: pre-training wins at full statistics** — with five-model ensembling the frozen-backbone arm is the *only* configuration in the study that passes both analysis gates, and both pre-trained arms show $\approx 3\text{--}4\sigma$ less residual structure than from-scratch. **(iii) Data efficiency: no advantage** — the pre-registered adoption metric, the closure left-shift, is undefined because neither fine-tuned nor from-scratch reaches the gates at *any* fraction, and the reduced-fraction ordering is mixed. In both (i) and (ii) the frozen arm matches or beats fine-tuning, locating the benefit in the pre-trained *representation* rather than in backbone adaptation. Section 7 documents three conversion errors found and fixed during analysis, one of which had inflated all closure numbers by a factor of ≈ 3 ; a fourth hypothesis, that EveNet’s diffusion-noised objective causes miscalibration, was refuted by source audit before the ablation was believed.

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1 Setup

Task. Learn per-event density ratios $r(x; \kappa_\lambda, \kappa'_\lambda)$ for an unbinned NSBI likelihood. The κ_λ dependence is a *continuous* deformation of one process’s spectrum (the m_{HH} lineshape and triangle–box interference), not the separation of two distinct processes.

Data. CMS 2024 full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ Powheg samples through the KIT CROWN chain (published $\mu\tau_h/e\tau_h$ selection), converted to EveNet’s stock 7-feature token schema: four tokens per event (b_1, b_2, τ_h, ℓ), missing transverse energy in the global conditioning block, weights = **genWeight** $\times$ **puweight** (signed; the reference carries 6.1% negative weights, the hypothesis 1.0%). Requiring a valid $H \rightarrow b\bar{b}$ candidate drops 18.6–24.5% of events (κ_λ -dependent). Class 0 = $\kappa_\lambda = 1$ (reference, 488k), class 1 = $\kappa_\lambda = 5$ (339k); merged, shuffled, split 80/10/10 into 661.9k train / 82.7k validation / 82.7k test. Training fractions multiply the *train* split; **all** runs share the one fixed test split.

Arms. **finetune** (20M pre-trained checkpoint, all weights trainable); **frozen** (same checkpoint, backbone GlobalEmbedding/PET/ObjectEncoder frozen, heads trainable — isolates the representation from optimiser effects); **scratch** (random init). Fractions $\{0.003, 0.01, 0.02, 0.05, 0.1, 1.0\}$ ($\approx 2.0\text{k}$ to 662k training events) $\times 5$ seeds = 90 runs. Options files are the authors’ own downstream recipes, vendored unchanged apart from the freeze flags.

External bar. A gradient-boosted decision tree on the 10 high-level features behind the published unbinned result (Sec. 6).

2 Pre-registration

Fixed before any sweep output existed, reproduced unchanged:

- (P1) **Decision primacy.** The adoption metric is the *closure left-shift*: the smallest fraction at which an arm passes both gates, compared between finetune and scratch; their ratio is the equivalent-data multiplier. A left-shift is an adoption-grade win *even at AUC parity*; an AUC-only win with failing closure is *not* a win. AUC is a secondary no-regression guard.
 - (P2) **Gates.** $|\text{IC}| \leq \max(1\%, 3\sigma_{\text{stat}})$ and $\text{SC}_{\text{rms}} \leq 0.05$ (= the analysis CLOSURE_TOL); $\chi^2/\text{ndf} \approx 1$ indicates noise-only residuals.
 - (P3) **Checkpoint selection.** Every arm is evaluated at its best-validation-loss checkpoint (EveNet’s predictor otherwise loads the final, most overfit, epoch).
 - (P4) **Fallback.** If no arm passes anywhere, leaving the left-shift undefined, the secondary reading is relative closure at matched fractions, reported as such.
- (P4) is the branch that materialised for the finetune–scratch comparison.

3 Metrics

3.1 The closure identity

The ratio estimate is $\hat{r} = \frac{p}{1-p} \cdot f_{\text{prior}}$, with f_{prior} fixed by the convention EveNet’s loss implies (Sec. 7). Everything rests on one identity: averaging the *true* ratio over reference events,

$$\mathbb{E}_{\text{ref}}[r] = \int p_{\text{ref}}(x) \frac{p_{\text{hyp}}(x)}{p_{\text{ref}}(x)} dx = \int p_{\text{hyp}}(x) dx = 1. \quad (1)$$

The reference density cancels and what remains integrates to one. Intuitively $r(x)$ says “this event is r times more likely under the hypothesis”; over the reference ensemble the $r > 1$ and $r < 1$ regions must balance exactly. On weighted Monte Carlo the expectation becomes the self-normalised weighted average $\sum_{i \in \text{ref}} w_i \hat{r}_i / \sum_{i \in \text{ref}} w_i$, converging at $1/\sqrt{n_{\text{eff}}}$ — so closure is always “consistent with 1”, never “equal to 1”.

3.2 The two metrics

Integral closure, IC = $\mathbb{E}_{\text{ref}}[\hat{r}] - 1$

Reweight the reference by $\hat{r}$ and ask whether the *total* hypothesis yield is recovered. IC = +0.20 means a 20% over-production. This is the *normalisation* error. Quoted with its weighted standard error.

Shape closure, SC_{rms} and SC_{norm}

IC can be satisfied by errors that cancel globally, so Eq. (1) is applied *differentially*: quantile-bin the score, and in each bin compare the $\hat{r}$ -reweighted reference mass A_b with the hypothesis mass B_b , forming $\text{rel}_b = A_b/B_b - 1$. We report the RMS over bins with each bin’s statistical variance subtracted in quadrature. We bin in the classifier score because it is monotone in $\hat{r}$, so miscalibration is forced to appear there.

Why the per-bin error is $1/n_{\text{eff}}$. For a weighted sum $S = \sum_i v_i$ with independent (Poisson) presence, $\text{Var}(S) = \sum_i v_i^2$, so

$$\frac{\text{Var}(S)}{S^2} = \frac{\sum_i v_i^2}{(\sum_i v_i)^2} \equiv \frac{1}{n_{\text{eff}}}, \quad n_{\text{eff}} \equiv \frac{(\sum_i v_i)^2}{\sum_i v_i^2}, \quad (2)$$

which *defines* the effective sample size (Kish). Unweighted events give $n_{\text{eff}} = N$ and the Poisson $1/\sqrt{N}$; unequal weights always reduce it ($v = (10, 1, 1, 1)$ gives $n_{\text{eff}} = 1.6$, four events behaving like 1.6). On the reference side $v_i = w_i \hat{r}_i$, so the reweighting inflates the spread — which is why bins at $\hat{r} \rightarrow \infty$ are dropped by the n_{eff} threshold rather than allowed to dominate. Since rel_b is a ratio of estimates from two *independent* samples, $\sigma_b^2 = 1/n_{\text{eff},0} + 1/n_{\text{eff},1}$.

Why subtract in quadrature. With $\text{rel}_b = \beta_b + \varepsilon_b$ (β bias, ε noise of variance σ_b^2), $\mathbb{E}[\text{rel}_b^2] = \beta_b^2 + \sigma_b^2$: the naive RMS is inflated by noise, worst at small statistics. Subtracting the known σ_b^2 estimates the bias RMS, $\text{SC}_{\text{rms}} = \sqrt{\max(\langle \text{rel}^2 - \sigma^2 \rangle, 0)}$. The clipping at zero is why exact 0.000 entries appear: they mean “no bias resolvable above the statistical floor”. Without the subtraction the metric rejects correct models — the first implementation did exactly that, failing an analytically calibrated toy in three seeds of four. The same σ_b normalises $\chi^2/\text{ndf} = \langle (\text{rel}_b/\sigma_b)^2 \rangle$.

3.3 SC_{rms} versus SC_{norm}

SC_{rms} is computed from $\hat{r}$ as delivered, SC_{norm} from $\hat{r}/\mathbb{E}_{\text{ref}}[\hat{r}]$. For a *pure* scale error $\hat{r} = c r_{\text{true}}$, every bin has $A_b/B_b = c$, so $SC_{\text{rms}} \simeq |c - 1| \simeq |\text{IC}|$ although the shape is perfect, while $SC_{\text{norm}} = 0$. The two raw numbers are therefore not independent, and reading them together gives three signatures:

signature	$ \text{IC} $	SC_{rms}	SC_{norm}	diagnosis
calibrated	≈ 0	≈ 0	≈ 0	usable
pure scale error	$ c - 1 $	$\approx \text{IC} $	≈ 0	normalisation only — removable
shape distortion	any	large	large	deformed likelihood — not removable

Table 1: Validated on synthetic data: a $\times 1.4$ scale error gives (0.399, 0.399, 0.000); a temperature-1.5 distortion gives (0.450, 0.694, 0.503).

Why the distinction is physical. Process p enters the likelihood as $\nu_p p_p(x)$ with ν_p the yield (from cross section $\times$ luminosity $\times$ acceptance, determined independently of the classifier) and $p_p = r_p p_{\text{ref}}$. If $\hat{r}_p = c r_p$ then $\nu_p \hat{p}_p = (c \nu_p) [\hat{p}_p / c]$: the constant is algebraically indistinguishable from mis-stating the *yield*, leaving the normalised shape untouched. It is therefore removable by construction (rescale by $\mathbb{E}_{\text{ref}}[\hat{r}]$ on held-out events) and, uncorrected, propagates only through that process’s normalisation where it is degenerate with a rate nuisance parameter. For a coupling measurement this is not harmless — $\sigma(\kappa_\lambda)$ carries the physics — but it is a one-parameter defect with a one-parameter cure. A shape error is different in kind: $\hat{r}(x)/r(x)$ varies with x , events are mis-weighted relative to one another, and no constant or rate parameter repairs it. SC_{rms} is the pre-registered gate; SC_{norm} is a post-hoc decomposition reported alongside.

What closure cannot catch. Both metrics test *self-consistency*, not *sufficiency*. A classifier on lossy features $\phi(x)$ converges to the exact ratio of the *marginalised* densities, satisfies Eq. (1) identically, and passes any test binned in a function of ϕ — the degenerate $\hat{r} \equiv 1$ passes both metrics perfectly while carrying no information. Information loss appears not as bias but as a *flatter* ratio, i.e. lost discrimination. Closure catches lies, not omissions: it earns $\hat{r}$ the right to enter the likelihood, while AUC shows it is worth entering.

4 Result 1 — discrimination (G0)

fraction	single model (seed mean $\pm$ s.d.)			5-seed ensemble		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.7427(57)	0.7498(80)	0.7573(122)	0.7553	0.7674	0.7698
0.01	0.7768(108)	0.7793(76)	0.7612(300)	0.7924	0.7914	0.7787
0.02	0.7941(43)	0.7931(80)	0.7804(19)	0.8040	0.8041	0.7845
0.05	0.8056(54)	0.8099(22)	0.7949(8)	0.8144	0.8160	0.7983
0.1	0.8163(22)	0.8169(9)	0.8003(25)	0.8210	0.8212	0.8059
1.0	0.8313(2)	0.8313(5)	0.8250(4)	0.8332	0.8330	0.8285

Table 2: Weighted AUC. Parenthesised digits are the seed standard deviation in units of the last decimals.

1. **Pre-training wins decisively.** At fractions 0.01–1.0 both pre-trained arms exceed scratch by 0.014–0.017 AUC against seed standard deviations of 0.001–0.008 — many standard deviations, monotone across the grid.
2. **Worth $\approx 3\text{--}4.5\times$ the data.** Interpolating scratch’s ensemble curve logarithmically, finetune’s AUC at 0.01, 0.02, 0.05, 0.1 requires scratch fractions of 0.033, 0.079, 0.222, 0.437 — multipliers 3.3, 4.0, 4.4, $4.4\times$; at full statistics scratch never reaches finetune’s value.
3. **Frozen $\approx$ finetune everywhere** (differences ≤ 0.004). The benefit lives in the pre-trained *representation*; adapting the backbone adds nothing measurable. This answers the “did the pretext retain κ_λ -relevant continuous information?” question: it did.

The 0.003 inversion. At the smallest fraction scratch leads. That tier is an optimisation regime, not a representational one: with $\approx 2\text{k}$ events and a global batch of $\approx 8\text{k}$ across 4 GPUs, one “epoch” is a single gradient step, and the selected checkpoints sit at epochs 4–21, i.e. the models are compared after $\mathcal{O}(10)$ updates under *different* learning-rate recipes. We treat it as a recipe artefact pending a batch-size/learning-rate ablation.

5 Result 2 — calibration (G0.5)

All numbers below use the ratio convention derived in Sec. 7. Production NSBI uses ensembles (the analysis config trains five members) and the BDT ceiling is itself an ensemble, so the **ensemble row is the like-for-like object**; single-model numbers are given for reference.

fraction	single model, IC (seed mean)			single model, SC_{norm}		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.136	0.310	0.164	0.328	0.476	0.365
0.01	0.110	0.279	0.213	0.346	0.341	0.206
0.02	0.148	0.120	0.129	0.276	0.244	0.187
0.05	0.102	0.099	0.158	0.178	0.206	0.159
0.1	0.098	0.143	0.192	0.176	0.168	0.180
1.0	0.050	0.053	0.055	0.090	0.061	0.076

Table 3: Single-model closure. Statistical errors on |IC| are 0.009–0.021. No single model passes the gates at any fraction.

arm	fraction	IC	gate	SC_{rms}	gate	verdict
finetune	1.0	0.0198	0.0559 ✓	0.0521	$0.05 \times$	fails by 4%
frozen	1.0	0.0527	0.0536 ✓	0.0272	0.05 ✓	PASSES
scratch	1.0	0.0362	0.0516 ✓	0.0742	$0.05 \times$	fails

Table 4: Ensemble closure at full statistics against the pre-registered gates ($|\text{IC}| \leq \max(1\%, 3\sigma_{\text{stat}})$, $\text{SC}_{\text{rms}} \leq 0.05$). No arm passes at any smaller fraction. χ^2/ndf : finetune 1.80, frozen 1.52, scratch 3.20.

(a) Ensembling works — once the convention is right. Because $\mathbb{E}_{\text{ref}}$ is linear in r , $\text{IC}_{\text{ens}} = \frac{1}{K} \sum_k \text{IC}_k$ identically: the ensemble reduces |IC| only if the members’ offsets differ in sign. Under the

superseded convention 15 of 18 cells were same-sign and ensembling did nothing. With the corrected convention the residuals straddle zero (several signed means are negative) and the ensemble collapses them: finetune at full statistics goes $0.050 \rightarrow \mathbf{0.0198}$ (statistical error 0.0186, i.e. consistent with zero), scratch at 0.01 goes $0.213 \rightarrow 0.068$.

(b) At full statistics the pre-trained arms are better calibrated in shape. Ordering at fraction 1.0 is consistent across all three shape measures: frozen < finetune < scratch on SC_{rms} (0.027/0.052/0.074) and on χ^2/ndf (1.52/1.80/3.20). With 20 bins, $\sigma(\chi^2/\text{ndf}) \approx \sqrt{2/20} = 0.32$, so scratch’s residuals are **3–4 σ** more structured than either pre-trained arm’s. **The frozen ensemble is the only configuration in the study that passes both gates** (Table 4).

(c) But there is no data-efficiency gain — the pre-registered metric is a null. (P1) compares finetune against scratch: neither passes at *any* fraction, so both first-passing fractions are undefined and no equivalent-data multiplier exists. Under the (P4) fallback the reduced-fraction ordering is mixed — scratch is nominally best on SC_{norm} at 0.01–0.05, the pre-trained arms at 0.1–1.0, and frozen is the *worst* arm at 0.003–0.01 while being the best at 1.0. There is no left-shift anywhere.

Summary. A usable density ratio requires the full 662k training events *and* ensembling. At that point pre-training measurably improves ratio shape; it does not reduce the data needed to get there.

6 The ceiling comparison

The feature-level BDT, measured at full statistics with class-normalised weights (so its prior factor is unity by construction):

fraction	AUC	IC	shape
0.01	0.7917	0.202	0.391
0.1	0.8130	0.011	0.039
1.0	0.8174	0.018	0.000

Table 5: Kinematic ceiling (10 published features, gradient-boosted trees).

Discrimination: EveNet wins. At 0.1 and 1.0 both pre-trained arms exceed the ceiling (0.821/0.821 vs. 0.813; 0.833 vs. 0.817), i.e. the object-level model extracts κ_λ information the 10 aggregate features discard. *Caveat:* the ceiling was measured on the un-cut flat-feature population whereas the sweep applies the valid- b -pair requirement, so the sweep population is slightly easier; the arm-to-arm comparisons share identical events and carry no such caveat.

Calibration: comparable, once conventions match. At full statistics the ceiling reaches $|IC| = 0.018$ and shape 0.000; the EveNet ensembles reach $|IC| = 0.020\text{--}0.053$ and $SC_{\text{rms}} = 0.027\text{--}0.074$. The frozen ensemble (0.053, 0.027) is within a factor of a few of the BDT on both axes. An earlier draft of this note reported the BDT as $\approx 8\times$ better calibrated *despite worse AUC* and drew a strong conclusion from it; that contrast was an artefact of converting the two estimators under different conventions (Sec. 7) and does not survive.

7 Corrections applied during analysis

Four hypotheses about the apparent calibration failure were tested. Three were our own conversion errors; the fourth was refuted before it was believed. Recorded because the first three each changed the numbers materially and the fourth would have cost 15 GPU jobs.

C1–C3: the ratio prior convention (a chain of three). EveNet’s classification loss is $\sum_i c_{y_i} w_i \text{CE}_i / \sum_i c_{y_i} w_i$, with $c = \text{normalization.pt}["\text{class_balance}"]$ and w the signed per-event weight (`apply_event_weight` is true). Its population optimum is $p/(1-p) = c_1 dW_1(x)/(c_0 dW_0(x))$, and since $r = [dW_1/W_1]/[dW_0/W_0]$,

$$r(x) = \frac{p}{1-p} \cdot \frac{c_0 W_0}{c_1 W_1}, \quad W_c = \text{class_counts}[c] \text{ (signed, train split)}. \quad (3)$$

We used, in order: W_0/W_1 from the test split (C1, wrong — omits c entirely); 1.0 (C2, “class-balanced” — correct only if c were inverse-frequency); and finally Eq. (3) (C3). The key fact is that `class_balance` is *inverse effective number of samples*, not inverse frequency — verified by reconstructing the stored vector bit-for-bit in `float32` from `class_counts` via $\beta = 1 - 1/N$ effective-number reweighting — so it cancels only 77.9% of the class imbalance in log space. Numerically $f = (1.2616 \times 11351.2)/(0.7384 \times 22569.1) = \mathbf{0.8593}$, against an empirically required 0.8702 ± 0.0498 measured over the 15 full-statistics runs (0.22σ ; no arm dependence). C2 inflated $|\text{IC}|$ by $\approx 3\times$ and, because a constant scale error shifts every bin equally, inflated SC_{rms} by the same amount — which is why the first pass showed closure *flat* in N rather than converging.

C4: refuted — the diffusion-noised objective. We hypothesised that EveNet’s classification head, trained with `noise_prob=1.0` but evaluated at the clean $t=0$ endpoint, is not a proper scoring rule there, and planned a 15-job ablation with `noise_prob=[0,0]`. Source audit refuted this before the result was believed: `noise_prob` is read at exactly one line, inside a `generation` branch that three flags in our own configs disable (`ReconGeneration.include: false, generation-recon: [0,0], classification-noised: [0,0]`). The branch that ran, `deterministic`, sets `full_time=0` — *identical* to predict time. There was no train/predict mismatch and the ablation would have been a no-op.

Other fixes. Closure is evaluated with signed weights (the measure the loss used); predict configs inherited an unresolvable relative `options.yaml` path and lacked the head-inclusion block, so every prediction task initially died before loading a checkpoint.

8 Caveats

- **No sensitivity claim.** No σ_{κ_λ} statement is made or implied; the stock schema discards τ -identification and FastMTT $m_{\tau\tau}$ (equally in all arms), and the MC-statistics estimator question is open upstream [3].
- **One channel, one hypothesis pair** ($\kappa_\lambda = 1$ vs. 5; μ_{τ_h}, e_{τ_h}). The $\kappa_\lambda = 0$ pair and the systematic-variation task are untouched.
- **Ensemble metrics have no error bar** beyond the statistical error on $|\text{IC}|$: each cell is a single ensemble. The χ^2/ndf comparison in Sec. 5(b) is the one shape statement with a quantified significance; the SC_{rms} ordering would need a bootstrap to be equally firm.

- **Frozen passes by a thin margin** ($|IC| = 0.0527$ against a gate of 0.0536). It should not be over-read as a robust pass.
- **Population mismatch vs. the ceiling only** (Sec. 6); all arm-to-arm comparisons share identical events.
- **The 0.003 tier** is optimiser-transient-dominated (Sec. 4).

9 Next steps

- N1. [P1, S] Bootstrap the ensemble closure metrics (resample test events) so the frozen-vs-scratch shape difference carries an error bar rather than a single number.
- N2. [P1, S] G1 corner-encoding A/B (τ_h at the vacant flag corner $(0, 0, \pm 1)$): the NPZs already exist for both encodings.
- N3. [P1, M] Money Plot 2 — systematic-variation data efficiency, using the analytic injector where closure is exact by construction. This is the regime the FM’s low-statistics claim was actually aimed at, and it is untested.
- N4. [P2, S] Low-fraction recipe ablation (batch size / learning rate at 0.003) to convert the AUC inversion from artefact to measurement.
- N5. [P2, S] Re-measure the ceiling on the valid- b -pair population to remove the asterisk in Sec. 6.

For the EveNet authors. (i) The ratio conversion of Eq. (3) is not documented anywhere in the repository and is easy to get wrong in three distinct ways (Sec. 7); for any likelihood-ratio use of the classifier head this is the single most valuable thing to document, or to expose as a utility. (ii) Our G1 result will be the first data on reusing *vacant flag corners* for object types absent from pre-training (τ_h, γ) — would they endorse this as the zero-surgery interim? (iii) Would they consider retrofitting an explicit type-embedding table, initialised from the current flag patterns so it is function-preserving on day one, making future object types an append-a-row operation rather than schema surgery?

References

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